

Lesson 7 Assessment

Secure Webhook (HMAC) — n8n Implementation Report

1. Assessment Summary

This document summarizes the implemented n8n webhook workflow for Lesson 7. The workflow receives POST webhook events, parses the incoming request, verifies an HMAC-SHA256 signature, and returns either HTTP 200 for a valid signature or HTTP 401 for an invalid signature.

2. Workflow Architecture

Step	Node	Purpose
1	Webhook Trigger	Receives POST requests at the custom endpoint lesson7-assessment.
2	Verify HMAC & Parse Payload	Reads headers/query/body and validates the HMAC-SHA256 signature.
3	Signature Valid?	Routes the request based on signature validation.
4A	Respond 200 OK	Returns HTTP 200 with an accepted status for valid requests.
4B	Respond 401 Unauthorized	Returns HTTP 401 when the HMAC signature is invalid.

3. Webhook Configuration

HTTP Method: POST

Custom Path: lesson7-assessment

Response Mode: response node

Raw Body: enabled for signature verification

The workflow is designed to receive event payloads and process them through the verification and routing logic before returning a response.

4. HMAC Security Implementation

The Code node uses HMAC-SHA256 to calculate an expected signature from the raw request body. It accepts an incoming signature from the x-signature or x-hub-signature-256 header, supports the sha256= prefix, and compares the supplied signature with the expected signature using a timing-safe comparison.

The implementation supports an n8n environment variable named WEBHOOK_HMAC_SECRET. A fallback placeholder secret is present in the workflow for environments where environment-variable access is unavailable.

5. Response Validation

Condition	HTTP Response	Expected Result
Valid HMAC signature	200 OK	Webhook accepted and processed
Invalid HMAC signature	401 Unauthorized	Webhook rejected

6. Payload Capture

The workflow extracts and returns the request headers, query parameters, and JSON body under the parsed object. A receivedAt timestamp is also generated for traceability.

7. Assessment Checklist

Requirement	Status
Webhook Trigger node configured	Completed
POST method and custom endpoint	Completed
Payload headers/query/body captured	Completed
HMAC signature verification	Completed
Valid request returns HTTP 200	Completed
Invalid request returns HTTP 401	Completed
Workflow execution validation	Demonstrated in n8n
Documentation	Completed

8. Challenges, Assumptions & Limitations

The HMAC secret must be shared consistently between the webhook sender and n8n. For a production deployment, the secret should be stored securely as an environment variable or credential rather than committed to workflow source code. The production webhook should be tested only after the workflow is activated.

9. Additional Enhancement

The workflow uses raw request body data when available, supports two common signature headers, and uses a timing-safe signature comparison to reduce timing-attack risk.

10. Source Workflow

The accompanying n8n JSON export contains the complete workflow definition, node configuration, routing, and HMAC verification logic.

Prepared for Lesson 7 — Working with Webhooks and Event-Driven APIs

Workflow: Lesson 7 - Secure Webhook (HMAC)